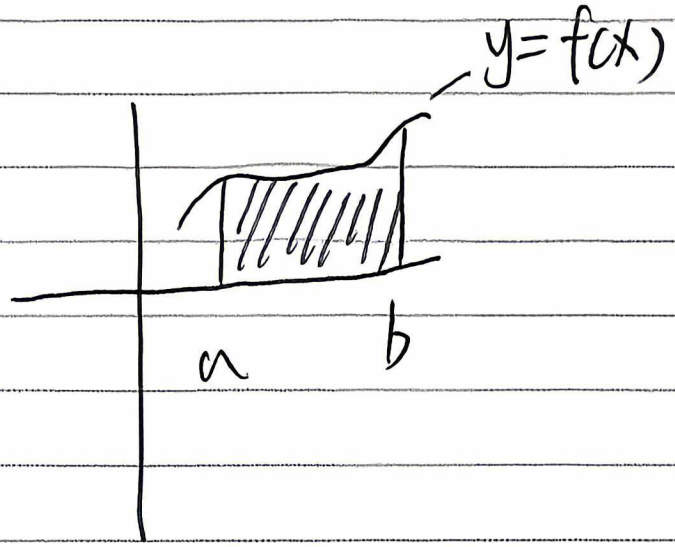
Integrals:

$$\int_a^b f(x)\,dx$$



example $y=x^2$

$$A=\frac{1}{n}\cdot f\left(\frac{1}{n}\right)+\frac{1}{n}\cdot f\left(\frac{2}{n}\right)+\cdots$$

$$\frac{1}{n}\cdot f\left(\frac{n}{n}\right)$$

$$=\frac{1}{n}\cdot\left(\frac{1}{n}\right)^2+\frac{1}{n}\cdot\left(\frac{2}{n}\right)^2+\cdots$$

$$\frac{1}{n}\cdot\left(\frac{n}{n}\right)^2$$

$$1^2+2^2+3^2+\cdots n^2=\frac{n(n+1)(2n+1)}{6}=\frac{1}{n^3}(1^2+2^2+3^2+\cdots n^2)$$

$$A=\frac{n(n+1)(2n+1)}{6n^3}=\frac{(n+1)(2n+1)}{6n^2}=\frac{1}{6}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)$$

$$\lim_{n\to+\infty} A=\lim_{n\to+\infty}\frac{1}{6}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)=\frac{1}{6}\times1\times2=\frac{1}{3}$$

$$A=\lim_{n\to+\infty}\left[f(x_1)\Delta x+f(x_2)\Delta x+\cdots+f(x_n)\Delta x\right]$$

Riemann Sum

$$\lim_{n \to +\infty} \sum_{i=1}^{n} f(x_i)\Delta x = \int_a^b f(x)\,dx$$

$$(\Delta x = \frac{b-a}{n} \quad x_i = a + i\Delta x)$$

$$\sum_{i=1}^{n} 1 = n \qquad \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6} \qquad \sum_{i=1}^{n} i^3 = \left(\frac{n(n+1)}{2}\right)^2$$

Fundamental theorem one

$$\int f(x)dx = F(x) \text{ then } \int_a^b f(x)\,dx = F(b) - F(a)$$

$$= \cancel{F(a)}\ F(x)\Big|_a^b$$

area above x-axis minus area below x-axis

~~pte~~ properties of integrals

① $\int_a^b (f(x)+g(x))dx = \int_a^b f(x)dx + \int_a^b g(x)\,dx$

② $\int c f(x)dx = c\int f(x)dx$ $c \to$ constant

③ $\int_a^b f(x)dx + \int_b^c f(x)dx = \int_a^c f(x)dx$

④ $\int_a^a f(x)dx = 0$

⑤ $\int_a^b f(x)dx = -\int_b^a f(x)dx$

⑥ if $f(x) \geq g(x)$ then

$\int_a^b f(x)dx \geq \int_a^b g(x)\,dx$

change of variables

$$\int_{u_1}^{u_2} g(u)\,du = \int_{x_1}^{x_2} g(u(x))\,u'(x)\,dx$$

only works u' does not change sign from $[u_1, u_2]$

$u = u(x)$ $\quad$ $u_2 = u(x_2)$

$du = u'(x)\,du$ $\quad$ $u_1 = u(x_1)$

example $\int_1^2 (x^3+2)^5 x^2\,dx =$

$u = x^3+2.$ $\quad$ $u_1 = 1^3+2 = 3$

$\frac{du}{dx} = 3x^2$ $\quad$ $u_2 = 2^3+2 = 10$

$du = 3x^2\,dx$ $\quad$ $\frac{1}{3}du = x^2\,dx.$

$$\int_1^2 (x^3+2)^5 x^2\,dx = \int_3^{10} u^5 \cdot \frac{1}{3}\,du = \frac{u^6}{18}\Big|_3^{10}$$

$$\Delta F = F(b) - F(a) = \int_a^b f(x)\,dx$$

$$\frac{\Delta F}{\Delta x} = \frac{1}{b-a}\int_a^b f(x)\,dx \quad (\text{Average } (f))$$